

School of Computing and Information Technology

Database Design CMP2018 (Practical)

Group Project – PART 2 (LAB ALPHA)

General Instructions

1. This lab is an individual assessment.
Although the ERD was completed in groups, each student must complete and submit their own SQL implementation of the assigned scenario. The ERD should guide your table structure, including primary and foreign key relationships.
 2. You are expected to demonstrate your understanding of data integrity rules through the proper use of constraints.
 3. The completed database must reflect realistic data entries consistent with the given scenario.
 4. Your SQL script must execute without errors in your SQL environment (SQLSERVER, MySQL, PostgreSQL, or Oracle).
 5. Include comments (--) to label each section of your script (e.g., -- CREATE TABLE, -- INSERT RECORDS).
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Task 1: Database Creation and Table Definition (15 marks)

- i) Write create and use statements for your database for your assigned scenario.
 - ii) Write CREATE TABLE statements for all entities in your ERD.
 - iii) Include all relevant constraints:
 - o Primary and foreign keys
 - iv) Demonstrate the implementation of at least one mandatory participation
 - v) Ensure to have at least one of each of the following:
 - o NOT NULL, UNIQUE, CHECK, and DEFAULT constraints.
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Task 2: Table Alteration (5 marks)

- Use ALTER TABLE statements to:
 - o Add a new column to an existing table.
 - o Modify the data type of an existing column.
 - o Drop a column
 - o Add a constraint

Task 3: Data Population (10 marks)

- Insert meaningful data into all tables in your database.
 - Include both **full** and **partial** inserts.
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Task 4: Data Modification and Deletion (4 marks)

- Use UPDATE statements to modify at least two existing data records (separate tables).
 - At least one update statement must either increase or decrease a value by a percentage.
 - Use DELETE statements to remove at least 2 records from separate tables while maintaining referential integrity.
 - At least one DELETE statement must have a criteria
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Task 5: Referential Integrity Demonstration (6 marks)

- Attempt to perform two separate operations that violates
 - Entity integrity; and
 - referential integrityand record the result or message in comments. Discuss your understanding of the result
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SUBMISSION REQUIREMENTS

Your submission must include:

1. The **SQL script file (.sql)** containing all implementation steps.
2. A **screenshot or exported output** showing that all statements executed successfully.
3. A brief **reflection paragraph** describing what you learned from implementing the ERD in SQL.

All submissions must be uploaded by the deadline announced by your tutor/lab technologist. Late submissions will be subject to penalties as outlined in the course policy.